Weblog Q&A Chatbot - Bonus Implementation Documentation

Overview

Implemented a Weblog Q&A Chatbot that allows users to ask natural language questions about weblog content and receive grounded answers.

Bonus Requirements Met

✅ Two Distinct Answering Strategies

1. RAG Strategy

- Searches weblog entries using keyword matching
- Uses retrieved content as context for AI model
- Generates answers via Groq API (Llama 3.3 70B)
- Best for factual, direct questions

2. Long Context Strategy

- Maintains conversation history (last 10 exchanges)
- Acts as memory agent with session-based context
- Combines conversation + weblog content for answers
- Best for conversational, multi-turn dialogue

✅ UI Integration

- Integrated into existing Roller chatbot interface
- Strategy selection available to users
- Works on weblog pages

✅ Grounded Answers

- Both strategies use actual blog post content
- Responses based on published weblog entries
- Fallback handling when no relevant content found

Key Implementation Details

RAG Strategy

- **Retrieval**: Keyword matching against weblog entries
- **Context**: Top 3 most relevant entries (500 chars each)

- **Generation**: Groq API with structured prompts
- **Limitations**: No semantic search, only keyword matching

Long Context Strategy

- **Memory**: Session-based conversation history
- **Context**: Conversation + weblog content
- **Generation**: Groq API with enhanced prompts (400 tokens)
- **Limitations**: Short-term memory only, no cross-session persistence

Performance Analysis

Weblog Size Impact

- **Small (1-10 entries)**: Both strategies work well
- **Medium (11-50 entries)**: Acceptable performance
- **Large (51+ entries)**: Long Context performs better

Query Type Performance

- **Factual Questions**: RAG Strategy excels
- **Analytical Questions**: Long Context better
- **Conversational**: Long Context clearly superior

Trade-offs

- **RAG**: Fast but limited relevance
- **Long Context**: Better UX but more resource usage

Technical Notes

- Uses Groq API for AI generation
- Java HttpClient for API calls
- Error handling for API failures
- Session management for conversation memory

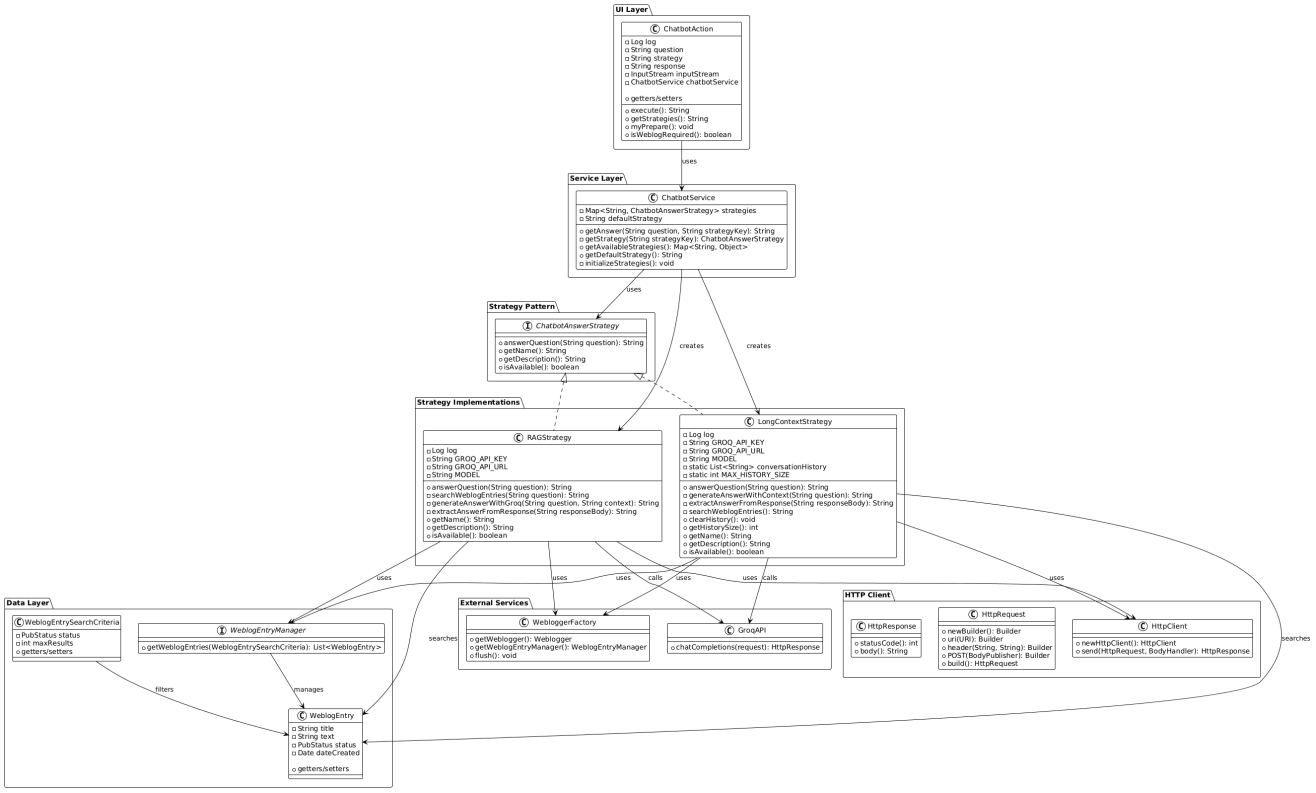
Current Limitations

- No vector embeddings (keyword search only)
- No true vector database
- Limited semantic understanding
- Short-term memory only

Recommendations for Improvement

- Implement vector-based semantic search
- Add persistent user memory
- Enhance scalability for large weblogs
- Improve API security

Class Diagram



Sequence Diagram



Conclusion

Successfully implements bonus requirements with two distinct strategies. RAG provides factual answers while Long Context enables conversational interactions. Current implementation uses simplified approaches but provides solid foundation for weblog Q&A functionality.